

DataSpy Decision AI

Key Decisions Report

Generated by: DataSpy Technologies

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Executive Decision Summary

This report presents the recommended actions generated by the decision engine. Each item includes business rationale, priority, and supporting evidence.

Decision Overview Table

Rank	Priority	Decision Type	Action
1	high	sales	Prioritize inventory, promotion, and planning around the strongest contributing products.
2	high	sales	Review concentration risk and explore diversification or protection strategies.
3	high	sales	Review demand assumptions and tighten short-term allocation until the pattern is clear.
4	medium	sales	Use the forecast for guided planning, not blind execution. Validate it again.

Detailed Recommendations

1. Focus attention on top-performing products

Recommended Action: Prioritize inventory, promotion, and planning around the strongest contributing products.

Priority: high

Decision Type: sales

Rank Order: 1

Rationale: A small group of products is contributing a meaningful share of revenue and deserves focused commercial and operational attention.

Business Explanation: A small group of products is contributing a meaningful share of revenue and deserves focused commercial and operational attention. Top-performing products are currently a key revenue concentration area.

What Happens If Ignored: The business may under-support the products that matter most and miss optimization opportunities in stock, promotion, or planning.

Confidence: high (91.4%)

Impact Estimate: high (90.0%)

Supporting Evidence:

```
{
  "insight_reference": {
```

```

"metric_type": "revenue",
"metric_column": "total_amount",
"top_products": [
  "Harrier",
  "Nexon",
  "Tiago",
  "Altroz",
  "Safari"
],
"top_5_total_value": 849899640,
"top_5_share_percent": 100,
"product_count": 5
},
"data_reference": {
  "product_column": "product",
  "value_column": "total_amount",
  "dataset_quality_label": "high",
  "dataset_quality_score": 0.9825
},
"schema_reference": {
  "product_role_confidence": 1,
  "value_role_confidence": 1
},
"capability_reference": {
  "sales": {
    "status": "enabled",
    "confidence": 0.6,
    "enable_reason": "minimum role coverage achieved; required core roles present",
    "role_breakdown": {}
  }
}
}
}

```

2. Reduce concentration risk across products

Recommended Action: Review concentration risk and explore diversification or protection strategies across the product mix.

Priority: high

Decision Type: sales

Rank Order: 2

Rationale: The top 5 products contribute 100.0% of total revenue, indicating a concentrated business base.

Business Explanation: The top 5 products contribute 100.0% of total revenue, indicating a concentrated business base. High dependence on a limited product set raises resilience concerns.

What Happens If Ignored: A change in demand for a small number of products could disproportionately affect overall business performance.

Confidence: high (91.4%)

Impact Estimate: high (90.0%)

Supporting Evidence:

```
{
  "insight_reference": {
    "metric_type": "revenue",
    "metric_column": "total_amount",
    "top_products": [
      "Harrier",
      "Nexon",
      "Tiago",
      "Altroz",
      "Safari"
    ],
    "top_5_total_value": 849899640,
    "top_5_share_percent": 100,
    "product_count": 5
  },
  "data_reference": {
    "product_column": "product",
    "value_column": "total_amount",
    "dataset_quality_label": "high",
    "dataset_quality_score": 0.9825
  },
  "schema_reference": {
    "product_role_confidence": 1,
    "value_role_confidence": 1
  },
  "capability_reference": {
    "sales": {
      "status": "enabled",
      "confidence": 0.6,
      "enable_reason": "minimum role coverage achieved; required core roles present",
      "role_breakdown": {}
    }
  }
}
```

3. Protect against short-term demand softening

Recommended Action: Review demand assumptions and tighten short-term allocation until the pattern becomes clearer.

Priority: high

Decision Type: sales

Rank Order: 3

Rationale: Forward-looking estimates indicate softer near-term movement, so aggressive allocation may create avoidable exposure.

Business Explanation: Forward-looking estimates indicate softer near-term movement, so aggressive allocation may create avoidable exposure. This helps prevent over-commitment when projected demand signals weaken. Forecast support: Available forecast signals point to a decreasing pattern over the forecast horizon with estimated movement of -37.65%. Reliability is marked as Medium and usability is usable with caution.

Forecast Support: Available forecast signals point to a decreasing pattern over the forecast horizon with estimated movement of -37.65%. Reliability is marked as Medium and usability is usable with caution.

What Happens If Ignored: The business may over-allocate stock, budget, or commercial effort into weaker demand conditions.

Confidence: medium (77.8%)

Impact Estimate: high (90.0%)

Supporting Evidence:

```
{
  "forecast_reference": {
    "forecast_mode": "predictive_forecast",
    "model_used": "seasonal_naive_7",
    "trend": "decreasing",
    "growth_percent": -37.65,
    "reliability_label": "Medium",
    "reliability_score": 0.65,
    "decision_usability_flag": "usable_with_caution",
    "validation_metrics": {
      "mae": 2677656.4,
      "rmse": 3153881.9893,
      "mape": 57.0363,
      "smape": 35.5693,
      "wape": 31.9921
    },
    "warnings": [
      "Duplicate timestamp ratio is 84.83%, which exceeds allowed threshold 10.00%.",
      "Target series shows moderate volatility.",
      "Validation error is high."
    ],
    "candidate_diagnostics": [
      {
        "model_name": "seasonal_naive_7",
        "mae": 2677656.4,
        "is_stable": true,
        "notes": [
          "stability_score=0.716",
          "consistency_score=0.0"
        ]
      }
    ]
  }
}
```

```

]
},
{
  "model_name": "drift_trend_model",
  "mae": 2913621.8867,
  "is_stable": true,
  "notes": [
    "stability_score=0.829",
    "consistency_score=0.0"
  ]
},
{
  "model_name": "seasonal_weekday_average",
  "mae": 3437581.6611,
  "is_stable": true,
  "notes": [
    "stability_score=0.816",
    "consistency_score=0.0"
  ]
},
{
  "model_name": "naive_last_value",
  "mae": 4454032.2,
  "is_stable": false,
  "notes": [
    "High variation across folds detected",
    "stability_score=0.615",
    "consistency_score=0.0"
  ]
},
{
  "model_name": "xgboost_lag_model",
  "mae": null,
  "is_stable": false,
  "notes": [
    "validation_failed=Feature frame is empty after feature engineering."
  ]
}
]
},
{
  "data_reference": {
    "dataset_quality_label": "high",
    "dataset_quality_score": 0.9825,
    "missing_percent_overall": 0,
    "duplicate_rows": 0
  },
  "capability_reference": {

```

```

"sales": {
  "status": "enabled",
  "confidence": 0.6,
  "enable_reason": "minimum role coverage achieved; required core roles present",
  "role_breakdown": {}
},
"inventory": {
  "status": "not_enabled",
  "confidence": 0.6,
  "enable_reason": "missing required roles: stock",
  "role_breakdown": {}
}
}
}
}

```

4. Apply forecast with caution

Recommended Action: Use the forecast for guided planning, not blind execution. Validate it against business context before major action.

Priority: medium

Decision Type: sales

Rank Order: 4

Rationale: A forecast is available, but reliability or usability indicators still require controlled usage.

Business Explanation: A forecast is available, but reliability or usability indicators still require controlled usage. Forecast signal exists, but reliability conditions still limit how strongly it should be used. Forecast support: Available forecast signals point to a decreasing pattern over the forecast horizon with estimated movement of -37.65%. Reliability is marked as Medium and usability is usable with caution.

Forecast Support: Available forecast signals point to a decreasing pattern over the forecast horizon with estimated movement of -37.65%. Reliability is marked as Medium and usability is usable with caution.

What Happens If Ignored: Large decisions may be made with more confidence than the current data and validation support.

Confidence: medium (66.1%)

Impact Estimate: medium (50.0%)

Supporting Evidence:

```

{
  "forecast_reference": {
    "forecast_mode": "predictive_forecast",
    "model_used": "seasonal_naive_7",
    "trend": "decreasing",
    "growth_percent": -37.65,
    "reliability_label": "Medium",

```

```
"reliability_score": 0.65,
"decision_usability_flag": "usable_with_caution",
"validation_metrics": {
  "mae": 2677656.4,
  "rmse": 3153881.9893,
  "mape": 57.0363,
  "smape": 35.5693,
  "wape": 31.9921
},
"warnings": [
  "Duplicate timestamp ratio is 84.83%, which exceeds allowed threshold 10.00%.",
  "Target series shows moderate volatility.",
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],
"candidate_diagnostics": [
  {
    "model_name": "seasonal_naive_7",
    "mae": 2677656.4,
    "is_stable": true,
    "notes": [
      "stability_score=0.716",
      "consistency_score=0.0"
    ]
  },
  {
    "model_name": "drift_trend_model",
    "mae": 2913621.8867,
    "is_stable": true,
    "notes": [
      "stability_score=0.829",
      "consistency_score=0.0"
    ]
  },
  {
    "model_name": "seasonal_weekday_average",
    "mae": 3437581.6611,
    "is_stable": true,
    "notes": [
      "stability_score=0.816",
      "consistency_score=0.0"
    ]
  },
  {
    "model_name": "naive_last_value",
    "mae": 4454032.2,
    "is_stable": false,
    "notes": [
```

```
"High variation across folds detected",
"stability_score=0.615",
"consistency_score=0.0"
]
},
{
"model_name": "xgboost_lag_model",
"mae": null,
"is_stable": false,
"notes": [
"validation_failed=Feature frame is empty after feature engineering."
]
}
],
"data_reference": {
"dataset_quality_label": "high",
"dataset_quality_score": 0.9825,
"missing_percent_overall": 0,
"duplicate_rows": 0
},
"capability_reference": {
"sales": {
"status": "enabled",
"confidence": 0.6,
"enable_reason": "minimum role coverage achieved; required core roles present",
"role_breakdown": {}
},
"inventory": {
"status": "not_enabled",
"confidence": 0.6,
"enable_reason": "missing required roles: stock",
"role_breakdown": {}
}
}
}
```